

BabyLM: what the 2025 scores looked like, and which evaluation bugs the 2026 pipeline fixed

Working note — explorations/babylm2025

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Abstract

This note does two things for a prospective 2026 BabyLM entrant. First, it audits — against the actual code of the 2026 evaluation pipeline (github.com/babylm-org/babylm-eval) — which of the concrete evaluation bugs reported in the 2025 challenge papers were fixed. Second, it shows what the spread of **BLiMP** and **(Super)GLUE** scores across 2025 systems actually looks like, so that a new score can be placed in context. The short story on the bugs: the BLiMP tie/NaN inflation and the age-of-acquisition mis-fit are fixed, the buggy WUG morphology tasks were deleted outright, but the EWok “concepts-absent-from-training” problem is a data issue that remains. The short story on the scores: BLiMP spreads widely (≈ 49 – 80) and is the metric that actually separates systems; (Super)GLUE is compressed (≈ 53 – 72) and barely distinguishes a $10\times$ difference in training data.

1 Sources and scope

The evaluation bugs are those documented in the 2025 BabyLM proceedings (`2025.babylm-main`; all 41 papers were read — see the companion `evidence.md`). The fix status is established by reading the 2026 pipeline source and its git history, not from any changelog (there is none). Paper references use the anthology number, e.g. (main.16). The score data are compiled from the primary submission papers; the caveats on that are stated in Section 3.

2 Which evaluation bugs were fixed in 2026?

Table 1 is the summary; the notes below give the mechanism and the code evidence.

BLiMP tie/NaN (fixed). In 2025, `rank_and_evaluate` chose the argmax candidate with `torch.max(...)[1]`, which returns the *first* index on ties and lets NaNs create spurious maxima; an order-invariant or degenerate model could therefore be scored as correct, and a dummy model could report 100.0. The 2026 code first maps NaN to $-\infty$ and then breaks ties *randomly*, so a model that cannot distinguish the minimal pair now scores at chance. This removes the inflation. It does *not* remove the 22 paradigms whose two sentences are permutations of one another — they are still in the data snapshot; they simply no longer inflate a degenerate model.

Age-of-Acquisition (fixed). The 2025 model-side sigmoid fit was unbounded and capped at `maxfev=1000`; on many clean learning curves the shape parameter diverged and the word was discarded, leaving the human–model correlation computed over a handful of survivors — exactly the “unpassed condition on the parameters of the fitted sigmoid” that main.29 diagnosed. The 2026 fit is *bounded*, uses $20\times$ more evaluations, averages surprisals within each checkpoint, corrects the chance-surprisal ceiling, and gates on a p -value. One caveat: the 2026 strict README reports AoA as an MSE-like quantity computed server-side, so the exact headline number may differ from the repository’s Pearson-correlation code; only the repo-side fit could be verified here.

WUG (removed). Rather than repair the morphological WUG past/adj tasks, 2026 deletes them: the `rank_and_evaluate_wug` routine, the task dispatch, and the collation entries are all gone. The

Issue (reported by)	2025 behaviour	2026 state	Verdict
BLiMP tie / NaN inflation (main.16)	<code>torch.max</code> picks first index on ties; NaNs propagate; a degenerate model can report 100.0	NaN $\rightarrow -\infty$ and <i>random</i> tie-breaking; degenerate models now score at chance (commit 9bcf1f2, Apr 2026)	Fixed
— underlying permutation items	22 of 67 BLiMP paradigms are word-order permutations	Same 67 paradigms still shipped; only the <i>scoring</i> changed	Partial
Age-of-Acquisition mis-fit (main.29)	Unbounded sigmoid fit (<code>maxfev=1000</code>) diverges and drops most words, so the correlation uses only 1–5 points	Bounded fit, <code>maxfev=20000</code> , per-checkpoint averaging, corrected chance ceiling, <i>p</i> -value gate (commit 7d0a6c8, Jul 2026)	Fixed
WUG evaluation bug (main.14); WUG=100 under morpheme tokenizers (main.21)	Buggy morphological WUG (past/adj) correlation tasks	Both WUG tasks <i>deleted</i> from the 2026 suite (commit 9bcf1f2)	Removed
EWoK at chance: $\sim 13\%$ of items use concepts absent from training (main.15)	Filter only drops out-of-vocab items; no training-coverage check	<code>ewok/dl_and_filter.py</code> and <code>vocab.txt</code> byte-identical; pair construction/collation fixed but coverage untouched	Not fixed
Reading-time task (fragility noted by main.17, main.31)	Surprisal–RT regression	Unchanged except a tokenizer-loading fallback	—

Table 1: Status in the 2026 pipeline of evaluation bugs reported in the 2025 challenge papers. “Verdict” colours: **Fixed** / **Partial** / **Removed** / **Not fixed**.

“WUG=100.00 under a morpheme tokenizer” artifact and the sign-flipping negative correlations are therefore moot — but so is any WUG signal.

EWoK (not fixed). main.15 showed EWoK sits at chance partly because $\sim 13\%$ of its test items involve concepts that never appear in the training corpus. That is a property of the *data*, and the 2026 filter and vocabulary files are byte-for-byte identical to 2025. Recent 2026 commits fixed how EWoK pairs are *constructed and scored* (two context-based minimal pairs; a prediction/target matching fix), which improves correctness, but the coverage problem — the reason EWoK barely rises above 0.50 — is untouched. Treat EWoK the same way in 2026 as in 2025: near-uninformative at this scale.

2.1 Other changes worth knowing

Task list. The 2026 strict zero-shot suite keeps BLiMP, BLiMP-Supplement, EWoK, Entity Tracking, COMPS, reading-time and AoA, *drops* both WUG tasks, and *adds* GlobalPIQA (length-normalised). Fine-tuning keeps the same seven (Super)GLUE tasks (BoolQ, MNLI, MRPC, MultiRC, QQP, RTE, WSC; MNLI/QQP subsampled to 10k). A new **Multilingual** track (English/Dutch/Chinese) adds MultiBLiMP, Hanzi structure/pinyin minimal pairs, and MECO reading-time.

Aggregation. Neither the 2025 nor the 2026 repository computes the headline Macro Average;

it is calculated server-side on the leaderboard. In 2025 it was the mean of the accuracy (“NLP”) average and the correlation (“human-likeness”) average — a 50/50 split. Whether 2026 keeps that exact weighting cannot be determined from the pipeline code, but the *composition* of the two halves has changed: the human-likeness side lost the WUG tasks, and the accuracy side gained GlobalPIQA.

3 What the BLiMP and (Super)GLUE spread looks like

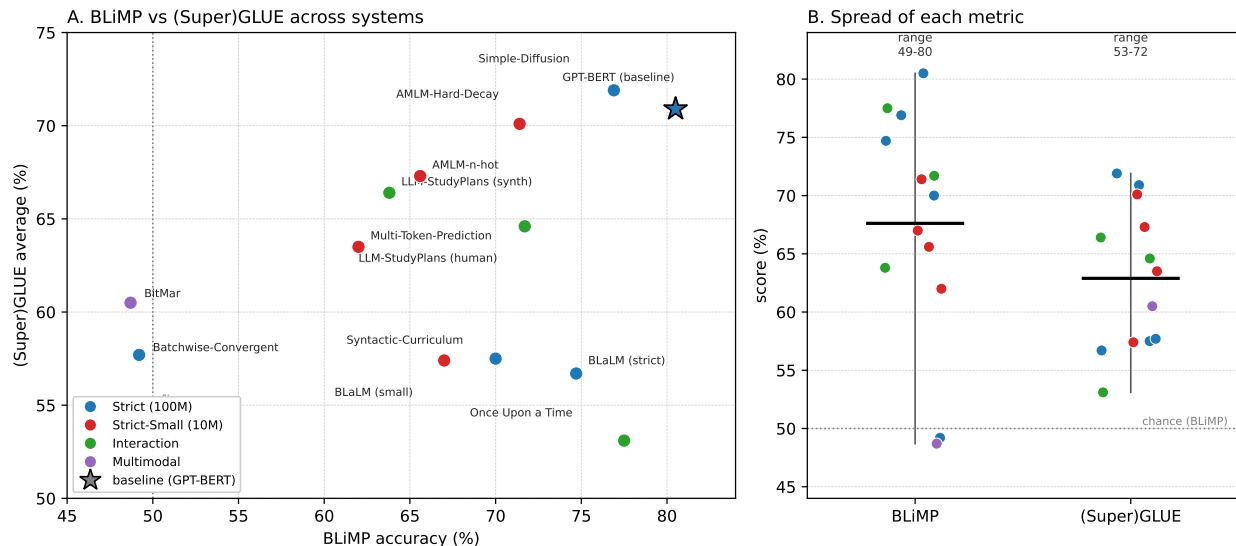


Figure 1: BLiMP versus (Super)GLUE for a cross-section of 2025 systems. **(A)** Each point is one system; colour is the track; the star is the official GPT-BERT baseline. **(B)** The same values shown as a one-dimensional spread per metric (black bar = mean, whisker = min–max). BLiMP ranges ≈ 49 –80; (Super)GLUE is compressed into ≈ 53 –72. Across systems the two are only weakly related (Pearson $r \approx 0.25$). Data compiled from the primary submission papers; each point’s BLiMP and GLUE come from the *same* source, but values are as-reported (including author re-implementations of the baselines) and GLUE-vs-SuperGLUE averaging conventions vary slightly. This is not the official leaderboard table, which is gated.

Three things stand out, and they are what matter for judging a new score.

BLiMP is the metric that separates systems. Its range across these systems is about 31 points (49 to 80), with a standard deviation near 9.5. A one-point BLiMP difference low down is meaningful; near the top it is not, because the strongest models approach the human/large-LM reference and the metric saturates.

(Super)GLUE is compressed and scale-insensitive. Its range is about 19 points (53 to 72), standard deviation near 6. The organisers’ own finding reinforces this: only two of seven Strict (100M-word) systems beat the best Strict-Small (10M-word) system on GLUE. A metric that barely reflects a $10\times$ data difference cannot finely rank models that differ by far less — so read GLUE as a coarse band, not a ranking.

The two are weakly correlated, so “good” is metric-specific. Systems with strong BLiMP and mediocre GLUE (Once Upon a Time: 77.5 / 53) and the reverse (Batchwise-Convergent: 49 / 58) both exist. A single system can look strong on one axis and ordinary on the other; the baseline

(GPT-BERT) is the rare point that is at the top of both.

3.1 Reference bands for placing your own score

From Figure 1 and the challenge results, the following anchors let a new score be judged without the leaderboard. They are approximate and track-dependent.

Metric	Chance / floor	Weak	Competitive	Strong / reference
BLiMP	50 (minimal pairs)	50–62	67–75	76–80 (GPT-BERT); 85–89 skyline
(Super)GLUE	$\approx$ 50 (majority)	53–57	58–66	67–72 (GPT-BERT $\approx$ 71)
EWoK	50	$\approx$ 50–54 for <i>everyone</i> — do not read differences		
Entity Track.	$\approx$ 20	high-variance (seed SD 6–9 pts); use only as a seeded probe		

Table 2: Rough reference bands for the metrics that carry signal, from the 2025 cross-section. “Skyline” for BLiMP is the human / large-LM reference reported by the organisers.

Practical reading: to know whether a submission is good, compare its **BLiMP** against the GPT-BERT baseline for its track (about 80 for Strict, high-60s to low-70s for Strict-Small) — that is the number that most cleanly says “ahead of” or “behind” the field — and treat **GLUE** as a wide competitive band around the high-50s to low-70s rather than a fine ranking. Do not use EWoK or the human-likeness tasks to decide whether a model is good; as the audit shows and the 2025 papers repeatedly found, small movements there are noise (and, for AoA in 2025, were partly a bug now fixed).

4 Bottom line

The 2026 pipeline is genuinely cleaner: the two most damaging scoring bugs (BLiMP tie/NaN inflation, AoA mis-fit) are fixed, and the unreliable WUG tasks are gone. The one substantive issue that survives is EWoK’s data-coverage problem, so EWoK remains near-uninformative. For contextualising a score, BLiMP is the axis to trust and to compare against the track baseline; (Super)GLUE is a coarse, compressed band; and the rest should be read with the scepticism the 2025 submissions themselves recommended.